

THE COLA-URINE DETECTIVE AGENCY — NEPHRITIC SYNDROME.

SUSPECT LINEUP

- SRIER bomb
- PSGN
- IgA
- HSP / IgA
- LUPUS
- ANTI-GBM
- ANCA
- ALP

WORKUP TOOLBOX

- BP
- ANCA
- ANTI-GBM
- ANTI-GBM
- CRYOGLOBULIN
- LEIACE
- NEPH/NEPHR
- STREPTOLYSIN
- ANCA
- BIOPSY

PENTAD.

- HEMATURIA + RBC CASTS + DYSMORPHIC RBCs
- SUBNEPHROTIC PROTEINURIA
- HYPERTENSION
- EDEMA — less than nephrotic
- AKI / OLIGURIA

GRAY AREA

OVERLAP — CAN BE NEPHROTIC OR NEPHRITIC

COMPLEMENT CLUE BOARD

↓C3 + NORMAL C4 (ALTERNATIVE PATHWAY)	↓C3 + ↓C4 (CLASSICAL PATHWAY)	NORMAL C3 + C4
PSGN	LUPUS	IgA
MPGN type II / C3G	CRYOGLOBULINEMIA	ANTI-GBM
atypical HUS	MPGN type I	ANCA
	POST-INFECTIVE ENDOCARDITIS	THIN BM
		ALPORT

RPGN — RAPID GFR DECLINE / EMERGENCY

- TYPE I — LINEAR IgG ANTI-GBM
- TYPE II — GRANULAR Immune Complex IgG, IgM, IgA, C3, C5b-9
- TYPE III — PAUCI-IMMUNE ANCA

URGENT TX — PLEX STEROIDS CYCLO PHOSPHAMIDE

1. Hematuria + RBC Casts / Dysmorphic RBCs

WHAT YOU SEE

Pentad, urine cups with casts

WHAT IT ENCODES

Hallmark of nephritic syndrome: glomerular inflammation lets red cells through, producing dysmorphic RBCs and RBC casts (pathognomonic of glomerular bleeding). Urine is cola/tea-colored. Distinguishes nephritic from nephrotic (bland sediment).

BOARD TRAP

RBC CASTS = glomerular origin; isolated non-dysmorphic hematuria without casts suggests urologic bleeding (stone, tumor), not GN.

2. Sub-nephrotic Proteinuria

WHAT YOU SEE

Pentad, moderate protein

WHAT IT ENCODES

Nephritic proteinuria is typically <3.5 g/day (sub-nephrotic), unlike the >3.5 g of nephrotic syndrome. Some overlap diseases (MPGN, severe lupus) can cross into nephrotic range.

BOARD TRAP

Proteinuria >3.5 g points toward nephrotic or an overlap lesion — pure nephritic syndrome stays sub-nephrotic.

3. Hypertension

WHAT YOU SEE

Pentad, blood pressure faces

WHAT IT ENCODES

Glomerular inflammation causes salt and water retention, expanding volume and raising blood pressure. A nephritic feature absent in pure nephrotic syndrome (where edema is oncotic). Treat with sodium restriction and diuretics.

BOARD TRAP

HTN + edema from volume overload is nephritic; nephrotic edema is from low oncotic pressure — the mechanism differs.

4. Edema (Periorbital)

WHAT YOU SEE

Pentad, puffy faces

WHAT IT ENCODES

Sodium/water retention produces periorbital and dependent edema. Unlike nephrotic anasarca, nephritic edema is volume-driven (overfill), accompanying hypertension.

BOARD TRAP

Periorbital edema + cola urine + HTN in a child after sore throat is the classic PSGN nephritic picture, not nephrotic syndrome.

5. AKI / Oliguria

WHAT YOU SEE

Pentad, kidney with AKI

WHAT IT ENCODES

Glomerular inflammation drops GFR, causing rising creatinine and oliguria. A rapid decline over days-weeks defines RPGN, a nephrologic emergency requiring urgent biopsy and immunosuppression.

BOARD TRAP

Falling GFR with active sediment over days = RPGN — don't wait; crescents on biopsy demand urgent treatment to prevent irreversible loss.

6. Suspect Lineup (Causes)

WHAT YOU SEE

Police lineup of culprits

WHAT IT ENCODES

Nephritic differential: PSGN, IgA nephropathy/IgA vasculitis, lupus, MPGN, anti-GBM (Goodpasture), ANCA vasculitis, and endocarditis-associated GN. Workup is organized by complement levels and serologies.

BOARD TRAP

Build the differential by complement + serology + IF pattern — pattern recognition (e.g., synpharyngitic vs post-infectious latency) narrows it fast.

7. Complement Clue Board

WHAT YOU SEE

Detective board, C3/C4 boxes

WHAT IT ENCODES

Key triage: Low C3 alone → PSGN, C3 glomerulopathy (alternate pathway). Low C3 + C4 → lupus, cryoglobulinemia, endocarditis (classical pathway). Normal complement → IgA, anti-GBM, ANCA vasculitis.

BOARD TRAP

This complement split is the highest-yield nephritic algorithm — normal complement does NOT mean mild disease (ANCA/anti-GBM are aggressive).

8. Gray Area — Overlap Diseases

WHAT YOU SEE

Board note, overlap

WHAT IT ENCODES

MPGN and diffuse proliferative lupus can present with mixed nephritic AND nephrotic features (heavy proteinuria plus active sediment and low complement). Don't force every patient into a single box.

BOARD TRAP

MPGN is the prototypic nephritic/nephrotic overlap with low C3 — a 'gray area' answer when both syndromes' features coexist.

9. RPGN — Rapid GFR Decline / Emergency

WHAT YOU SEE

Red ER room banner

WHAT IT ENCODES

Rapidly progressive GN = nephritic syndrome with crescents on biopsy and rapid renal failure. Three immune categories: anti-GBM (linear IF), immune-complex (granular, e.g., lupus/PSGN), and pauci-immune ANCA. Treat urgently with steroids, cyclophosphamide/rituximab, +/- plasmapheresis.

BOARD TRAP

All three RPGN types show crescents — the IF pattern (linear vs granular vs pauci) tells you which one and guides whether plasmapheresis (anti-GBM) is needed.

10. Anti-GBM RPGN

WHAT YOU SEE

ER bed, anti-GBM placard

WHAT IT ENCODES

Type I RPGN: linear IgG along GBM, often with pulmonary hemorrhage (Goodpasture). Anti-collagen IV antibodies. Plasmapheresis is essential to remove pathogenic antibody plus immunosuppression.

BOARD TRAP

Anti-GBM is the RPGN where plasmapheresis is most clearly indicated — don't omit it while waiting for steroids to work.

11. ANCA / Pauci-immune RPGN

WHAT YOU SEE

ER placards, ANCA

WHAT IT ENCODES

Pauci-immune crescentic GN with normal complement; PR3/c-ANCA (GPA) or MPO/p-ANCA (MPA, EGPA). Often with systemic vasculitis (sinusitis, lung, neuropathy). Treat with steroids + rituximab/cyclophosphamide.

BOARD TRAP

Pauci-immune means scant IF staining — a normal-complement RPGN with negative immunofluorescence is ANCA until proven otherwise.

12. Workup Toolbox

WHAT YOU SEE

Open toolbox of tests

WHAT IT ENCODES

Essential nephritic workup: urinalysis with microscopy (casts), complement (C3/C4), ASO/anti-DNase B, ANA/anti-dsDNA, ANCA, anti-GBM, cryoglobulins, hepatitis B/C, blood cultures (endocarditis), and renal biopsy when unclear or rapidly progressive.

BOARD TRAP

Order the serologic panel BEFORE empiric immunosuppression — and biopsy early in RPGN; the IF/serology combination, not urine alone, makes the diagnosis.

13. Cola-Colored Urine

WHAT YOU SEE

Detective's cup of dark urine

WHAT IT ENCODES

Tea/cola-colored urine reflects glomerular hematuria with degraded hemoglobin, a visual signature of nephritic syndrome. Combined with RBC casts it localizes bleeding to the glomerulus.

BOARD TRAP

Cola urine with RBC casts = glomerular; cola urine WITHOUT casts could be myoglobinuria (rhabdo) or hemoglobinuria — check sediment and dipstick-vs-microscopy mismatch.

14. Microscopy / RBC Casts Confirmation

WHAT YOU SEE

Detective's microscope

WHAT IT ENCODES

Phase-contrast/microscopic exam confirming dysmorphic RBCs and RBC casts is the single most useful bedside test to confirm glomerular origin and trigger the nephritic workup.

BOARD TRAP

Dysmorphic RBCs/acanthocytes confirm glomerular bleeding — isomorphic RBCs point to a urologic source needing cystoscopy/imaging instead.